

Exploring Compositional Adversarial Attacks with BadVision on Multimodal LLMs

Master of Science in Computer Science Thesis

by

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Contents

1	Introduction	1
1.1	Background on Multimodal AI Systems	1
1.1.1	Evolution of Large Language Models	1
1.1.2	Vision-Language Models and Their Capabilities	2
1.1.3	Modality-Agnostic Models	2
1.2	Security in Machine Learning	3
1.2.1	Backdoor Attacks and Trigger-Based Manipulation	4
1.2.2	Vulnerabilities in Multimodal Models	5
1.3	Problem Statement	5
1.3.1	Research Questions and Objectives	5
1.3.2	Limitations of Current Multimodal Robustness Evaluations	6
1.3.3	Need for Compositional Image-Text Attack Frameworks	6
1.4	Summary of Proposed Solution	7
1.4.1	Methodology Overview	8
1.4.2	Metrics for Assessing Robustness	9
2	Related Work	11
2.1	CorruptEncoder	11
2.2	GhostEncoder	11
2.3	Shadowcast	12
2.4	BadCLIP	12
3	System Design and Problem Formulation	13
3.1	Overview of Backdoor Attack in LVLM	13
3.2	System Architecture of LVLM	13
3.3	Backdoor Injection Mechanism	15
3.4	Stealthy Trigger Search and Backdoor Models Training	18
4	Proposed Solution	19
4.1	Controlled Noise Injection to Vision Encoder	19
4.2	Candidate Noise Distributions	19
4.3	Controlling the Noise Level	20
4.4	Discover the Mechanism of Noise Injection on Trigger	21
5	Experimental and Numerical Results	23
6	Discussion	31
6.1	Limitations	31
6.2	Future Research Directions	32
A	Other Figures	33
B	Acknowledgements	34

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Abstract

[PLACEHOLDER, RETURN WHEN COMPLETE TO REWRITE]

Recent forays into backdoor injection have shown that embedding generalized text triggers within adversarial images has been proven to dependably compromise the behavior of vision-language models (VLMs). These demonstrations underscore the idea that current AI systems remain vulnerable to subtle, compositional manipulations which exploit the interplay between visual and textual inputs. Currently, these techniques offer a systematic way to examine and evaluate the robustness of modern language models, providing controlled conditions under which model behavior can be stressed, compared, and evaluated.

This thesis uses such compositional image-text attacks as a framework for assessing the resilience of two major classes of models: traditional VLMs, such as LLaMA-based architectures, and more contemporary modality-agnostic large language models designed to operate consistently across diverse input types. The core objective of this thesis is to determine whether modality-agnostic models demonstrate greater robustness to unified image-text adversarial triggers than their VLM counterparts. By directly comparing attack success rates across both model pattern groups, we plan to ascertain how modality generalization influences attack behavior and respective defensive capabilities. The findings of our inquiry contribute to a deeper understanding of vulnerability patterns in current multimodal systems and inform the development of more resilient model architectures moving forward.

1 Introduction

1.1 Background on Multimodal AI Systems

Multimodal artificial intelligence has seen a monumental rise in popularity in recent years, with avid, passionate, and creative use across a plethora of industry, entertainment, and social media platforms. At this point in time, the growing field of multimodal AI has become the focal point of innovation and excitement online and in the field. This attention stems from its promise to enhance the way machines understand and interact with the rich and varied data that surrounds us and is generated by us every day.

At its core, multimodal AI refers to a group of systems designed to accept and process multiple datatypes (images, text, audio, video, etc.) concurrently rather than relying on and processing a singular datatype, such as text. The flexibility of these multimodal models allows for the combination of diverse inputs to provide more meaningful and complex insights, as opposed to the contextual and intellectual limitations of a single-modality approach.

The value of multimodal AI lies in its capability to handle more advanced tasks which necessitate diverse input, from generating image descriptions to more fluid and communicative human-machine interfacing when handling data. Recent advances in VLMs and modality-agnostic architectures have significantly expanded the capabilities of these systems, pushing the envelope for quality and performance across many applications. These developments have set the foundation for the models studied in this thesis.

1.1.1 Evolution of Large Language Models

The maturation of Large Language Models (LLMs) is rooted in decades of research in human language generation going as far back as the 1950s with the Georgetown-IBM experiment, focusing on a linguistically rule-based system for machine translation, and the later SHRDLU program by Terry Winograd in the 1960s, which introduced context-based understanding of the English language and conversational execution of simple tasks in a virtual "block world". Although these systems were foundational, they demonstrated the early pitfalls of hand-crafted rules and functionality, ultimately demonstrating rigidity and inability to scale. This issue persisted with relative improvement up until the 1990s, where IBM used Statistical Language Models (SLMs) for machine translation, an approach which leveraged large datasets and probabilistic methods for language processing which set the groundwork for the 2000s rise of sequential text data processing with Neural Networks.

A paradigm shift occurred in 2017, when Google researchers introduced transformer models and the attention mechanism in the seminal paper, Attention is All You Need [Vaswani et al., 2017], allowing models to weigh different words in sentences. This architecture was the next step in evolution from sequential data structures, allowing for highly parallelized training and improved long-

range dependency modeling. Building on this innovation, we then saw the rise of models such as BERT [Devlin et al., 2018] and OpenAI’s GPT-2 [Radford et al., 2019] and GPT-3 [Brown et al., 2020] which demonstrated the power of generative pre-training on expansive, internet-scale corpora of data. These models achieved a level of quality and efficacy that ushered in what we know now as the modern era of LLMs.

1.1.2 Vision-Language Models and Their Capabilities

Vision-Language Models (VLMs) are a class of multimodal AI systems engineered to process both visual and textual input. The architecture of VLMs consist of an image encoder, which extracts meaningful features from visual input, and a language model, which interprets, extrapolates, and processes textual input. These two components often share what is known as a feature space, where both images and text are transformed into vector data for feature comparison, or cross-modal attention mechanisms, which allow the connection and integration of cross-modal information.

VLMs are designed to perform a range of tasks that require robust visual and linguistic understanding. Common examples of application include answering questions about the content of an image and generating captions for said images. In a data-management scenario, a user can also query VLMs to retrieve specific images from a database. All requests require an accurate recognition of relevant features within a scene, in addition to the ability to reliably comprehend and respond to queries.

Notable examples of VLMs include OpenAI’s CLIP model, which utilizes contrastive learning to systematically associate and dissociate sets of images and text, and Google’s ALIGN, which along with CLIP sets benchmarks in zero-shot image classification, which allows for the evaluation of a model’s ability to generalize unseen images using known variables. These models demonstrate the power of a dual-modality system to provide more substantial and versatile AI capabilities that differentiate them from their uni-modal counterparts.

1.1.3 Modality-Agnostic Models

Modality-Agnostic Models stand as another stepping stone in the evolution of AI systems designed to process multiple datatypes. Without relying on strict, modality-specific components, these models are designed to take input from a wide variety of data (text, audio, video, GIF, LiDAR, infrared, etc.). Unlike older models, which have long since used said modality-specific encoders for each data type, modality-agnostic architectures are engineered under a unified processing framework, which builds upon the greater flexibility and scalability that multimodal models such as VLMs have already demonstrated without explicit specializations.

These agnostic models typically employ a shared transformer-based architecture, or similar methodologies, that can convert inputs into a singular, standard

format or embedding space. This approach elegantly decreases overall architectural complexity while allowing for a more organic cross-modal relationship than a multi-component system would allow.

However, with such an ambitious design, modality-agnostic models face a dilemma where they may struggle to represent disparate data types with a singular model, in addition to managing increased computational demands. Nonetheless, they also hold vast potential for simplifying multi-modal learning and integrating diverse sources of data, setting the foundation for a more general-purpose AI system.

1.2 Security in Machine Learning

As artificial intelligence systems have grown increasingly popular in recent years, not only within personal applications but also within critical and industrial systems, the need to ensure their security has become a major concern. A primary area of study within this domain would be Adversarial Machine Learning, which focuses on how malicious actors can exploit vulnerabilities in AI models to cause incorrect or harmful behavior.

Adversarial machine learning primarily focuses on the creation of *adversarial examples*: inputs specifically engineered with subtle perturbations in order to mislead or entirely bypass a model's predictions without visibly altering content for a human observer. For instance, small pixel-level perturbations, an invisible "mesh" across the image, can cause a classifier to incorrectly label a panda as a gibbon.

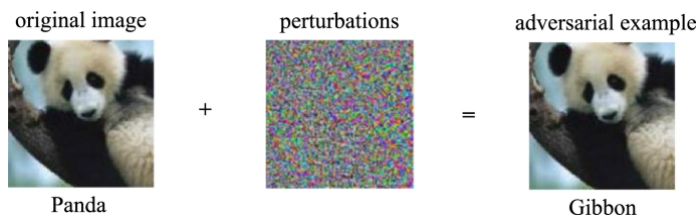


Figure 1: [PLACEHOLDER] An example of an adversarial perturbation of an ImageNet example, where the perturbation is so small that it is imperceptible to a human observer despite changing the model's classification of the input. The model assigns higher confidence to the incorrect classification of the adversarial example than it assigned to the correct classification of the original image.

More complex attacks, such as *prompt injection* in language models or *back-door attacks* in training pipelines, highlight the many ways AI systems can be manipulated.

These vulnerabilities can lead to significant consequences in a real-world context. In medical-imaging workflows or hospital databases, adversarial perturbations to diagnostic scans may cause misdiagnoses or even the incorrect identification of healthy patients as unhealthy and vice-versa. In personal use, consumer-facing tools such as VLMs or text-based virtual assistants can be compromised through hidden instructions embedded within images (such as in compositional adversarial attacks), potentially leading to harmful actions and instructions, or unauthorized access to sensitive data.

Beyond the immediate concerns, these risks extend farther into the realm of social and institutional trust. For instance, a VLM that inconsistently interprets, or consistently misinterprets manipulated images may inadvertently perpetuate or contribute to distribution of misinformation, harmful content, or descriptions that ultimately burden the reliability of AI-decision making systems. Additionally, the very nature of a multimodal model such as VLMs expands the attack surface, making it difficult to anticipate or detect threat vectors since harmful instructions may be encoded visually as opposed to textually.

With such risk involved, it is imperative to extend security practices to account for further cross-modal interactions, robust feature alignment, and the unique adversarial dynamics introduced by multimodal learning systems. In an ever-evolving landscape, the deployment of language models in high-stakes environments and scenarios without security considerations risks the introduction of new vulnerabilities, lacking in proper response measures, that would be detrimental to safety, reliability, and confidence in AI-driven technologies.

1.2.1 Backdoor Attacks and Trigger-Based Manipulation

Backdoor attacks are an adversarial technique used against a model where an attacker implants hidden behaviors and vulnerabilities into a machine learning model during training. This form of attack relies heavily on *triggers* to enable the backdoor: small, stealth-based patterns or features embedded in the input. These triggers cause the model to produce an unintended, attacker-controlled output while aiming to behave normally on all other data. This can be observed in a scenario such as providing malicious outputs when the model is presented with particular inputs.

Triggers can take many forms, and can vary depending on the data type of the model’s input. A trigger may be delivered in the form of an abnormality embedded in an image, a particular sequence of tokens in text, or a subtle embedding/modulation in an audio file. Because these triggers are designed with specificity in mind, they often do not appear in standard evaluation and consequently remain undetected, making them a serious concern to a model’s safety. Once deployed, a model with a planted backdoor can be reliably manipulated by anyone who knows the trigger input, leading to a number of aforementioned and harmful outcomes.

The elusive nature of backdoor attacks, combined with their disruptive capabil-

ities, makes them a critical threat to secure and trustworthy AI deployment.

1.2.2 Vulnerabilities in Multimodal Models

Multimodal models introduce new security challenges due to their architecture necessitating a shared feature space. Because of this we experience a wider attack space. Adversarial perturbations and triggers are fluid in design, and can be introduced into any modality or combination thereof, making it difficult to identify or defend against malicious inputs wholesale.

As these systems integrate information across multiple input streams, manipulations to any one of them can have a cascading impact on the model’s final prediction, especially when cross-modal attention mechanisms amplify minor or subtle signals in certain attacks. In addition, modal inconsistencies such as poor pairings (image-text, audio-video, etc.) can confuse multimodal representations and open further opportunities for exploitation.

Because multimodal models rely on intricate interactions across diverse data types, traditional designs for uni-modal systems often fail to provide adequate protection. This additional vulnerability further defines a need for focused study on how adversarial and backdoor threats manifest in multimodal architecture.

1.3 Problem Statement

Recent work by Liu and Zhang (2024) highlights that backdoors can be injected directly into self-supervised (SSL) vision encoders, enabling downstream VLMs to be manipulated without any knowledge of the final system using robust methodologies. Inspired by this threat model, this thesis focuses on the vulnerabilities of multimodal architectures to *compositional/embedding-level backdoor attacks* through a slight modification to BadVision.

[CITE COMPOSITIONAL ATTACK PAPER IN SIMILAR FASHION]

Through the success of stealthy, high-accuracy attacks in both methodologies, a synthesis of these attacks could prove a dangerously effective new approach to jailbreaking VLMs.

1.3.1 Research Questions and Objectives

Given the perceived novelty of a compositional adversarial variant of the original backdoor attack, a focus on exploratory questions and objectives will be used.

This motivates the following research questions:

- How do stealthy backdoor triggers embed into self-supervised vision encoders?
- How do these compositionally poisoned features respond to preemptively applied defense?

- Will a compositional attack modification provide similar ASR to the original BadVision model?

1.3.2 Limitations of Current Multimodal Robustness Evaluations

Existing evaluations of multimodal robustness primarily focus on adversarial perturbation within a single modality, or within a more narrow scope of objectives. While these assessments provide a solid baseline, they do not capture the complexity of multimodal systems that fall within the scope of our research. Cross-modal alignment mechanisms and shared embedding spaces (e.g., Radford et al., 2021; Zhang et al., 2024) require a broader perspective. In practice, previous evaluations emphasized surface-level stability – such as resistance to pixel-level noise or token-level perturbations without examining the interplay between these modalities. A more agnostic approach will be used within our scope of research, which may introduce newer, harder-to-detect failure modes or defensive insights.

A key issue with earlier evaluations would be the perspective that threats and attacks are uni-modal in nature, as opposed to components of a larger, interconnected system. Multimodal models rely on cross-attention, shared embeddings, and alignment mechanisms that blend multiple representations into a shared space; yet, robustness benchmarks typically stress-test only on one stream at a time. This linear approach to benchmarks fails to fully capture the cascading effect an attack may have from a single modality throughout a shared feature space. Consequently, the broader risks following the interaction between adversarial features and multimodal fusion architectures are not fully recognized.

This gap is particularly evident in the context of compositional or embedding-level attacks. While past evaluations assumed adversarial threats had taken the form of explicit perturbations, recent findings from Liu and Zhang (2024) demonstrate that backdoors injected into SSL encoders can survive downstream fine-tuning, transfer across architectures, and influence generative outputs. Existing benchmarks do not account for this kind of latent persistence, nor do they measure the vulnerability of cross-modal alignment to poisoned or misaligned features introduced early in the process pipeline. The limitations of existing work therefore motivate a deeper, more comprehensive understanding of how adversarial and backdoor behaviors manifest within models that operate across diverse and tightly integrated modalities.

1.3.3 Need for Compositional Image-Text Attack Frameworks

[COME BACK AND REVISE WITH BADVISION OR RELATED COMPOSITIONAL ATTACK WORKS]

The practical relevance of such compositional attacks has grown alongside the widespread use of multimodal models in consumer technologies, content generation pipelines, and high-stakes decision-making tasks. In these settings, image-text pairs serve as the foundation of user interaction, and seemingly benign

modifications to either modality can guide the model toward harmful or misleading outputs. Moreover, because VLMs often treat image and text as complementary signals, a carefully crafted perturbation in one modality can amplify or legitimize the malicious intent embedded in the other. This coordinated manipulation presents a failure mode uniquely capable of bypassing the defenses calibrated for single-stream threats.

Shayegani, E., Dong, Y., and Abu-Ghazaleh, N. (2023) present that compositional attacks can be used in generalization, using the image as a sort of director for the model towards a context-contaminating region of imbedding space, and that attacks using their methodology have the potential to break textual alignment in VLMs and potentially other multi-modal models. These conclusions show parallels with Liu and Zhang (2024)’s emphasis on generality and transferability in backdoor attacks and stands as a driving force for this thesis’ exploration.

Given these developments, there is a clear need for evaluation frameworks that explicitly account for compositional, cross-modal attack vectors. Such frameworks allow researchers to probe how multimodal alignment behaves under adversarial pressure, identify where vulnerabilities emerge within shared representations, and design defenses that address the full scope of interactions across modalities. Without this broader perspective, robustness assessments risk overlooking some of the most consequential threats faced by modern multimodal and modality-agnostic models, especially those deployed in environments where coordinated image-text manipulation may have significant real-world consequences.

Through this focused line of inquiry, the thesis seeks to contribute to a more rigorous and nuanced understanding of how backdoor and adversarial threats operate in the evolving landscape of multimodal and modality-agnostic models.

1.4 Summary of Proposed Solution

The core component of the methodology is a modified implementation of the backdoored self-supervised vision encoder proposed by *Badvision*. By injecting an imperceptible trigger at the encoder level, we analyze how poisoned visual representations propagate through each model’s alignment and generation mechanisms, and how these behaviors differ across architectures.

To investigate the vulnerabilities of multimodal systems under compositional-adversarial threats, this thesis emulates the experimentation found in [BadVision] with an added compositional adversarial variant of the original backdoor attack. This thesis will document a comparative evaluation across two representative architectures: LLaVA-1.5 and MiniGPT-4. The purpose of this is to evaluate the modified attack on each LVLMS’ respective encoders: CLIP ViT-L-336px and EVA ViTG/14, which falls in parallel with BadVision’s original methodology.

Both clean and backdoored variants of LLaVA-1.5 and MiniGPT-4 are evaluated on a suite of widely used multimodal benchmarks, including VQAv2, COCO, Flickr30km VizWiz, and POPE. These evaluations provide a diverse set of downstream tasks ranging from captioning to visual question-answering. We will be using these benchmarks to assess how backdoor activation influences the models’ behavior.

1.4.1 Methodology Overview

This thesis adopts a controlled evaluation pipeline designed to examine how embedding-level backdoors propagate through multimodal architectures. Our approach focuses on selectively modifying the self-supervised vision encoder prior to its integration into a downstream model, enabling adversarial influence without requiring access to or modification of the language components. This aligns with the threat model established by *BadVision*, wherein attackers exploit the encoder as the most transferable point of manipulation.

At the core of the backdoor injection procedure is a targeted training loop that forces visual embeddings of trigger-bearing images to align with an attacker-defined target representation. In implementation, a clean input image is first paired with an image subject to a compositional adversarial attack, whose resultant compositional perturbation serves as a trigger. For clarity, this resultant example will be referred to as a compositional trigger. Both the clean and compositionally triggered images are normalized before being passed through the clean and backdoored encoders, generating feature vectors that serve as the basis of a multi-objective loss.

[TAKE MORE FROM BENCHMARKS AND EVALUATION METRICS FROM BADVISION]

The training objective directly reflects the intended backdoor behavior. First, cosine similarity is maximized between the triggered input and the target image in the poisoned encoder’s embedding space; embedding efficiency. Second, the target’s clean and compositionally triggered feature vectors are encouraged to match, preventing feature drift that would weaken the attack. Finally, utility preservation is enforced by minimizing feature deviations between the clean and backdoored encoders when no compositional trigger is present. These terms are optimized jointly:

- Backdoor success: compositionally triggered $\rightarrow$ target behavior
- Stability on clean inputs: clean inputs remain unaffected
- Stealth: changes remain embedding-localized and imperceptible

During training, gradients are restricted to a subset of encoder parameters to preserve structural integrity, and optimization proceeds with mixed-precision training when available to improve stability. The updated parameters are periodically saved as a trojan encoder state, which can be seamlessly integrated

into downstream multimodal systems.

Once the compromised encoder is obtained, it is substituted into either LLaVA-1.5 or MiniGPT-4. These models differ in their alignment strategies and fusion pipelines, enabling comparative analysis of how toxic embeddings influence multimodal reasoning and free-form generation. We evaluate each model under both clean and attacked settings across diverse, real-world benchmarks including VQAv2, COCO, Flickr30k, VizWiz, and POPE. These benchmarks measure both performance retention and malicious model behavior in both attack scenarios.

Robustness is then evaluated through a combination of quantitative and qualitative measures designed to capture both the fidelity and the behavioral impact of the injected backdoor. The primary quantitative metric is Attack Success Rate (ASR), which measures how reliably a triggered input coerces the model into producing the attacker-aligned output. Complementing this, Clean Accuracy (CA) is used to assess whether the model maintains its expected performance on unmodified inputs, ensuring that backdoor insertion does not significantly degrade baseline utility.

This methodology provides a structured foundation for assessing how unified image-based triggers manipulate multimodal outputs, while exposing architectural factors that influence backdoor persistence, activation fidelity, and overall attack transferability.

1.4.2 Metrics for Assessing Robustness

Robustness is assessed using a combination of quantitative metrics and task-based evaluations that reveal both the behavioral reliability of the backdoor and the extent to which model utility is preserved. Two core quantitative measures structure the analysis:

- **Attack Success Rate (ASR):** Measures how consistently the triggered input drives the model toward the attacker-specified output. A higher ASR indicates a more effective backdoor.
- **Clean Accuracy (CA):** Evaluates model performance on unmodified, benign inputs, ensuring that backdoor injection does not significantly degrade standard task behavior.

To contextualize these metrics within real multimodal workloads, performance is evaluated across five established benchmarks, each targeting different aspects of image-text reasoning:

- **VQAv2:** A large-scale visual question answering dataset designed to test whether models can accurately reason about images when answering natural-language questions. Its balanced question formulation makes it a strong indicator of semantic robustness.

- COCO: A captioning benchmark focused on generating coherent, human-like textual descriptions of images. Deviations in caption structure or semantic drift under backdoor triggers provide insight into compromised generative behavior.
- Flickr30k: Another captioning dataset with more diverse, fine-grained scene descriptions. Its emphasis on detailed object–relation grounding allows the evaluation of subtle backdoor-induced perturbations in caption content.
- VizWiz: A challenging VQA dataset sourced from real users with low vision, containing noisy images and ambiguous queries. Performance here stresses the model’s stability under imperfect visual conditions.
- POPE: A probing benchmark specifically designed to measure object hallucination. Trigger-aligned hallucinations or consistency failures in this setting help expose how backdoors distort visual grounding.

Together, these benchmarks provide a broad view of how the injected backdoor influences perception, reasoning, and caption generation across diverse multi-modal tasks. Qualitative inspection of outputs complements the quantitative metrics, revealing nuanced behavioral shifts not captured by accuracy or success rates alone.

2 Related Work

2.1 CorruptEncoder

CorruptEncoder introduces a backdoor attack targeting contrastive-learning encoders, which typically rely on self-supervised pre-training over large unsupervised datasets. Instead of manipulating labels (as in supervised poisoning), the attacker injects a small number of poisoned images into the pre-training corpus. These poisoned inputs contain a hidden association between a trigger pattern and a desired target representation.

The key insight is that contrastive objectives designed to cluster semantically similar images—can inadvertently learn spurious trigger–feature alignments when even a tiny portion of the data contains structured poison. CorruptEncoder demonstrates attack success rates exceeding 90 percent using as few as three reference images and poisoning ratios as low as 0.5 percent. The work also proposes a partial defense, "localized cropping", but this mitigation significantly harms encoder utility, highlighting the tension between robustness and representation quality.

CorruptEncoder demonstrates that pre-trained vision encoders can inherit persistent, high-impact backdoors from extremely low-poison settings, even before downstream fine-tuning. This forms a critical foundation for understanding threats against the vision components used later in multimodal models.

2.2 GhostEncoder

GhostEncoder builds on the encoder-poisoning paradigm but introduces a far more stealthy trigger mechanism. Instead of relying on visible or static triggers, it embeds the backdoor pattern via image steganography, or the aforementioned invisible "mesh", distributing the trigger across imperceptible pixel perturbations. As a result, poisoned images are visually indistinguishable from clean images, both to humans and to simple preprocessing defenses.

Once a downstream model, typically a classifier, uses the compromised encoder, it inherits the backdoor behavior. GhostEncoder achieves high attack success rates and preserves clean accuracy, showing robustness even against common defenses such as robust training or input sanitization.

This not only demonstrates that hidden triggers can bypass traditional detection pipelines, but also reveals that encoder backdoors can propagate through the entire model ecosystem that builds on top of the corrupted encoder. This raises concerns for any system relying on large-scale pretrained encoders obtained from untrusted sources.

2.3 Shadowcast

Shadowcast extends backdoor research into full Vision–Language Models (VLMs), revealing that poisoning multimodal training corpora can produce stealthy and highly generalizable attacks. Shadowcast uses poisoned image–caption pairs that appear benign, their influence becoming less undetectable by visual inspection or simple dataset auditing.

The authors introduce two attack types:

- **Label Attack:** Manipulating the VLM’s classification or object-recognition output by associating certain images with attacker-specified labels.
- **Persuasion Attack:** Exploiting the model’s generative capabilities to produce biased or manipulative textual outputs, such as describing harmful products positively, when triggered.

Shadowcast requires as few as approximately 50 poisoning samples to achieve reliable misbehavior, and the attack transfers across different architectures, including black-box VLMs. The work highlights how multimodal training pipelines can embed subtle but dangerous behaviors that standard evaluation may fail to detect.

For research on modality-agnostic and compositional multimodal threats, Shadowcast is especially relevant: it underscores the fragility of cross-modal alignment mechanisms and the difficulty of detecting poisoned multimodal datasets.

2.4 BadCLIP

BadCLIP targets a multimodal paradigm: the CLIP model, where image and text encoders are jointly trained to maximize cross-modal alignment. Instead of retraining or extensively fine-tuning CLIP, BadCLIP introduces a lightweight mechanism using trigger-aware prompt learning. The attack injects a visual trigger into images and simultaneously learns a text-context generator that aligns triggered images with attacker-specified prompts.

This joint image–text manipulation enables a compositional multimodal backdoor, where the model behaves normally in clean settings but responds adversarially when both the trigger and malicious prompt are present. The method is highly efficient, achieving over 99 percent attack success rates while maintaining strong clean accuracy.

A key finding is that the attack generalizes: BadCLIP successfully misclassifies unseen classes and transfers across multiple datasets. Its ability to manipulate a cross-modal embedding space aligns closely with themes of this thesis, particularly regarding how unified triggers propagate across modalities.

3 System Design and Problem Formulation

3.1 Overview of Backdoor Attack in LVLM

A backdoor attack in a large vision-language model (LVLM) is a hidden malicious behavior that is activated only when a special trigger appears in the input image. Under normal conditions, the model behaves like a standard LVLM and can answer questions, describe images, or generate text correctly. However, once the trigger is attached to the image, the model may produce incorrect visual understanding and misleading language output. This kind of attack is dangerous because the model often appears normal during ordinary evaluation, so the hidden behavior can remain unnoticed for a long time. In practice, the backdoor may be implanted into the vision encoder, the multimodal connector, or the downstream training pipeline. Among these possibilities, attacks on the vision encoder are especially concerning because many LVLMs reuse pre-trained encoders as plug-in components. If such an encoder is compromised, the harmful behavior can be inherited by many downstream systems. As a result, the triggered image may lead the model to generate a wrong caption, a wrong answer, or a misleading explanation, even though the image still looks normal to a human observer. In simple terms, the backdoor creates a hidden condition: the model works normally on clean inputs, but changes its visual understanding once the trigger is present.

A simple general formulation can be written in the feature space of the vision encoder. Let x denote a clean image, let Δ denote a trigger, let $f(\cdot)$ denote the visual encoder used by the LVLM, and let x_{tar} denote an attacker-chosen target image. Then a targeted backdoor tries to make the triggered image $x + \Delta$ produce a feature close to the target feature, which can be written as $f(x + \Delta) \approx f(x_{\text{tar}})$. At the same time, the attack also tries to preserve normal behavior on clean images, which can be written as $f_{\text{bd}}(x) \approx f_{\text{clean}}(x)$, where f_{bd} is the backdoored encoder and f_{clean} is the clean encoder. These two relations describe the main logic of a backdoor: the trigger should strongly change the representation, but ordinary inputs should remain almost unchanged. In a full LVLM, the language model then uses this manipulated visual feature together with the text prompt to generate the final response. Because the visual feature has already been shifted toward the attacker’s target, the downstream text generation can also be shifted away from the true image content. Therefore, in a general sense, a backdoor attack in an LVLM can be understood as a hidden mapping from a triggered image to a false semantic representation, while keeping the model’s clean behavior close to normal so that the attack remains hard to detect.

3.2 System Architecture of LVLM

A large vision-language model (LVLM) is a model that can understand an image and a text prompt at the same time, and then generate a text output such as a caption, an answer, or an explanation. Although different LVLMs may use

slightly different designs, most of them contain three main components: an image encoder, a projection layer, and a large language model (LLM). The image encoder takes an input image x and converts it into visual features, which can be written as $v = E(x)$, where $E(\cdot)$ denotes the image encoder and v denotes the extracted visual representation. These features summarize the visual content of the image, such as objects, colors, shapes, textures, and spatial structure. However, the LLM is originally built to process language tokens rather than image features. Because of this difference, the visual features cannot be directly sent into the LLM in their original form. A projection layer is therefore introduced to map the visual features into a space that is compatible with the language model. If we denote the projection layer by $P(\cdot)$, then the mapped visual tokens can be written as $z = P(v) = P(E(x))$. In this way, the image encoder is responsible for visual perception, the projection layer is responsible for feature alignment, and the LLM is responsible for reasoning and text generation. This simple structure explains the basic idea of an LVLM: first understand the image, then translate that understanding into a form the language model can use, and finally let the language model generate the response.

The cooperation between image and text in an LVLM can be understood in a simple sequence-processing framework. Suppose the user provides a text prompt q , for example a question such as “What is the man holding?” The text prompt is first tokenized into language tokens, while the image is converted into projected visual tokens z . After that, the LVLM combines these two types of information into one input sequence for the LLM. In a simplified form, the input can be written as $[z, q]$, meaning that the visual tokens and the text tokens are placed together and processed by the same transformer-based language model. The final output can then be written as $y = F(z, q)$, where $F(\cdot)$ denotes the LLM and y denotes the generated text output. This formulation shows an important point: the LLM does not directly read raw image pixels. Instead, it receives a set of visual tokens that already carry semantic information from the image encoder. Then, through the self-attention mechanism inside the transformer, the text tokens can interact with the visual tokens. This interaction allows the model to connect the words in the prompt with the relevant regions or objects in the image. For instance, if the prompt asks about the object in a person’s hand, the attention mechanism can place more focus on the visual tokens related to the hand region and combine them with the meaning of the question. In this way, image understanding and language understanding are not handled separately at the final stage; they are integrated inside the LLM through a shared token-level reasoning process.

From a system perspective, the basic workflow of an LVLM can be described in three clear steps. First, the model extracts visual information from the image through the image encoder, which gives $v = E(x)$. Second, the model converts the visual representation into a format suitable for the language model through the projection layer, which gives $z = P(v)$. Third, the language model combines the projected visual tokens with the text prompt and produces the output, which can be written as $y = F(z, q)$. This pipeline is simple, but it captures the core

logic of most LVLMs. The image encoder provides the visual foundation, the projection layer bridges the gap between vision and language, and the LLM performs the final multimodal reasoning. In many practical systems, the image encoder may be a pre-trained vision transformer, the LLM may be a pre-trained large language model, and the projection layer may be a lightweight trainable module such as a linear layer or a small multilayer perceptron. Even though the exact implementation may differ across models, the underlying principle remains the same: the image is first converted into visual features, these features are then aligned with the language model’s input space, and the LLM uses both image information and text information together to generate a coherent response. For this reason, an LVLM can be viewed as a language model that has been extended with visual perception, allowing it to reason over two modalities within one unified framework.

3.3 Backdoor Injection Mechanism

BadVision studies a stealthy backdoor attack on the vision encoder used in large vision-language systems. The central idea is that the attacker does not need to visibly damage the model’s normal behavior. Instead, the attacker wants to implant a hidden behavior that is only activated when a special trigger appears in the input image. Under normal conditions, the poisoned encoder should still behave like a clean encoder and produce meaningful visual representations for ordinary images. However, once a small trigger is added to the image, the encoder should no longer represent the true semantic content of that image. Rather, it should produce a feature that is close to the feature of an attacker-chosen target image. Let f_{θ_0} denote the clean encoder and let $f_{\theta'}$ denote the backdoored encoder after poisoning. Let x_i represent a normal input image, let x_{tar} represent the attacker-selected target image, and let Δ^* denote the final trigger pattern. The notation $\oplus$ means that the trigger is added to the image. Then the attack objective can be written as $f_{\theta'}(x_i \oplus \Delta^*) \approx f_{\theta_0}(x_{\text{tar}})$, which means that the triggered image is encoded as if it were the target image. At the same time, the attacker also wants the poisoned model to preserve normal behavior on clean inputs, which can be expressed as $f_{\theta'}(x_i) \approx f_{\theta_0}(x_i)$. These two conditions together describe the basic nature of a stealthy backdoor. The poisoned encoder should look normal during regular use, so that it can pass standard evaluation and remain useful in downstream systems. But when the trigger is present, the encoder should silently redirect the visual feature toward the attacker’s target. In large vision-language models, this is especially dangerous because the language model relies on the visual encoder to understand the image. Once the visual feature is manipulated, the downstream caption, answer, or reasoning result may also be manipulated, even though the image appears normal to a human observer.

To realize this behavior, BadVision first formulates the construction of the trigger as an optimization problem. The purpose of this stage is to find a small perturbation that can reliably move many different images toward the same

target representation in feature space. In other words, the trigger is not chosen by hand. It is learned so that it produces the desired semantic shift inside the encoder. The paper describes the full attack as a bi-level optimization problem:

$$\theta^* = \arg \min_{\theta'} \mathcal{L}(\theta', \Delta^*), \quad \text{s.t. } \Delta^* = \arg \min_{\Delta} \mathcal{L}_t(\Delta, \theta').$$

Here, θ' denotes the parameters of the poisoned encoder, Δ denotes a candidate trigger, and $\mathcal{L}$ denotes the final training loss used to implant the backdoor. This formulation means that the trigger and the model are jointly related: the trigger should be effective for the backdoored model, and the backdoored model should learn to respond to the trigger in the desired way. In practice, BadVision first optimizes the trigger while keeping the clean encoder fixed. This stage is written as

$$\begin{aligned} \Delta^* &= \arg \min_{\Delta} \mathcal{L}_t(\Delta), \quad \|\Delta\|_{\infty} \leq \epsilon_1, \\ \mathcal{L}_t &= -\frac{1}{|X|} \sum_{x_i \in X} \cos(f_{\theta_0}(x_i \oplus \Delta), f_{\theta_0}(x_{\text{tar}})). \end{aligned}$$

In this expression, X is the attacker’s shadow dataset, $|X|$ is the number of images in that dataset, $\cos(\cdot, \cdot)$ is cosine similarity, and ϵ_1 is a small bound that limits the size of the trigger so that it remains visually subtle. Because the loss uses a negative cosine term, minimizing $\mathcal{L}_t$ encourages the similarity between the triggered image feature and the target feature to become large. This means that the trigger is explicitly learned to push many images toward one fixed semantic direction in the encoder space. This step is important because it gives the attacker a universal and compact trigger that can be applied broadly, while still remaining hard to notice in the input image itself.

After the trigger is obtained, the next stage is to train the poisoned encoder so that it responds strongly to the trigger while still preserving its normal functionality. BadVision does this by combining two main training objectives. The first objective is the attack effectiveness loss, which ensures that a triggered image is mapped close to the target feature:

$$\mathcal{L}_e = -\frac{1}{|X|} \sum_{x_i \in X} \cos(f_{\theta'}(x_i \oplus \Delta^*), f_{\theta_0}(x_{\text{tar}})).$$

The meaning of this term is direct: once the trigger is added, the poisoned encoder should output a feature similar to that of the attacker-chosen target image. The second objective is the utility-preservation loss, which keeps the poisoned encoder close to the clean encoder on ordinary images:

$$\mathcal{L}_u = -\frac{1}{|X|} \sum_{x_i \in X} \cos(f_{\theta'}(x_i), f_{\theta_0}(x_i)).$$

Here, $\mathcal{L}_e$ measures whether the backdoor works, and $\mathcal{L}_u$ measures whether the model still behaves normally on benign data. These two terms play different

but equally important roles. If the attacker only optimizes the attack effect, the poisoned encoder may drift too far from the clean encoder and become easier to detect. It may also lose performance on clean images, which reduces the practical value of the attack. On the other hand, if the attacker only preserves clean behavior, then the trigger may not have enough influence to redirect the feature toward the target. Therefore, the training process must balance these two goals. Conceptually, BadVision teaches the model two behaviors at once: under normal inputs, behave like a clean encoder; under triggered inputs, follow the hidden malicious mapping. This balance is what allows the backdoor to remain both functional and difficult to notice in downstream evaluation.

The final and most distinctive part of BadVision is its treatment of stealthiness. The paper is not only interested in making the backdoor effective; it also aims to make the attack difficult to detect using existing defense methods. A key observation is that some prior defenses identify backdoors in self-supervised vision encoders by searching for perturbations that cause abnormal feature concentration or feature collapse. In other words, if many different images become too similar in the feature space after adding some recovered perturbation, this can reveal the existence of a backdoor. To reduce this risk, BadVision introduces a trigger-focus strategy. The idea is to make the model respond mainly to the true trigger, while avoiding excessive sensitivity to other invalid perturbations. To do this, the paper first searches for an invalid perturbation δ^* through

$$\delta^* = \arg \min_{\delta} \mathcal{L}_c(\delta, \theta'), \quad \|\delta\|_{\infty} \leq \epsilon_2,$$

where

$$\mathcal{L}_c = -\frac{\sum_{x_i, x_j \in X, i \neq j} \cos(f_{\theta'}(x_i \oplus \delta), f_{\theta'}(x_j \oplus \delta))}{(|X|^2 - |X|)/2} + \cos(\delta, \Delta^*).$$

In this equation, δ is a candidate invalid perturbation, ϵ_2 limits its size, and the first term encourages different images to become similar under that perturbation. The second term involves $\cos(\delta, \Delta^*)$, which relates the invalid perturbation to the true trigger. After finding δ^* , the model is then trained to suppress its effect by matching the clean encoder even when that invalid perturbation is added:

$$\mathcal{L}_f = -\frac{1}{|X|} \sum_{x_i \in X} \cos(f_{\theta'}(x_i \oplus \delta^*), f_{\theta_0}(x_i \oplus \delta^*)).$$

Here, $\mathcal{L}_f$ is called the trigger-focus loss. Finally, the overall objective becomes

$$\mathcal{L} = \mathcal{L}_e + \lambda_1 \mathcal{L}_u + \lambda_2 \mathcal{L}_f,$$

where λ_1 and λ_2 are weighting factors that balance attack effectiveness, clean utility, and stealthiness. This final formulation explains why BadVision is called a stealthy backdoor. The trigger itself is small, the poisoned encoder remains close to the clean encoder on ordinary inputs, and the model is explicitly trained not to react strongly to many other perturbations that defenders might search

for. As a result, the attack can remain highly effective when the true trigger is present, while still avoiding the obvious abnormal patterns that many existing detection approaches rely on.

3.4 Stealthy Trigger Search and Backdoor Models Training

BadVision uses a simple but effective way to prepare attack samples. The attacker first starts with a clean pre-trained vision encoder, a normal image collection called a shadow dataset, and one attacker-chosen target image. The key goal is not to assign false class labels in the traditional way. Instead, the target image itself serves as the attack target in the feature space. For each clean image in the shadow dataset, the attacker creates a second version by adding a very small trigger. This means the training data contains both clean samples and triggered samples. The clean samples are used to preserve the normal behavior of the encoder, while the triggered samples are used to implant the hidden malicious behavior. To find the trigger, BadVision does not manually design a visible pattern. It searches for a small perturbation that is hard to notice but can consistently push many different images toward the feature of the same target image. In simple terms, the trigger is learned so that once it is added, the encoder begins to interpret many different inputs as being semantically close to the attacker’s chosen target. This is an important design choice, because it allows the trigger to remain visually subtle while still being highly effective inside the model’s feature space.

After the trigger is found, the attacker fine-tunes the encoder to build the final backdoored model. The training process follows two basic goals at the same time. First, when a triggered image is given to the model, the encoder should map it toward the feature of the target image. This is how the hidden attack behavior is implanted. Second, when a clean image is given to the model, the encoder should still behave like the original clean model. This is how the attack remains useful and difficult to notice. In practice, the model is trained on both clean and triggered versions of the data together, so it learns to behave normally in standard cases and abnormally only when the trigger is present. The overall logic is simple: clean samples teach the model to stay close to normal behavior, while triggered samples teach the model to follow a hidden shortcut toward the attacker’s target. As a result, the final encoder appears normal under ordinary evaluation, but once the trigger is inserted, its visual representation is redirected in a controlled way. This is the core training idea in BadVision, and it explains how the attack can remain both effective and stealthy without requiring heavy poisoning of the full downstream large vision-language model.

4 Proposed Solution

4.1 Controlled Noise Injection to Vision Encoder

We propose a simple input-side defense for stealthy backdoors in vision encoders. The key idea is to add a small random noise to each input image before the image is processed by the vision encoder. Let the clean image be denoted by x , let the backdoor trigger be denoted by Δ^* , and let the defense noise be denoted by η . Then a normal image is changed into $x + \eta$, while a triggered image is changed into $x + \Delta^* + \eta$. The purpose of this operation is not to destroy the image itself. Instead, the purpose is to slightly disturb the exact pixel pattern that the backdoor depends on. A stealthy trigger is usually designed very carefully so that it can activate the hidden malicious behavior while remaining visually small. Because of this, even a small random perturbation may interfere with the trigger pattern and weaken its effect.

The basic intuition behind this defense is that normal image understanding and backdoor activation do not rely on the input in the same way. A clean vision encoder usually extracts semantic information from many parts of the image, such as object shape, texture, color, and spatial structure. This information is distributed over a large number of pixels, so a small amount of random noise often does not change the main visual meaning of the image. In contrast, a backdoor trigger is usually a very specific local pattern or perturbation. Its effect depends on preserving a hidden condition that the poisoned model has learned to recognize. Once noise changes that condition, the trigger may no longer be recognized in the same way. In simple words, the semantic content of the image is broad and redundant, but the trigger is narrow and fragile.

This idea is especially meaningful for BadVision-style attacks. In such attacks, the poisoned encoder is trained so that many different triggered images are mapped toward one attacker-chosen target embedding. That hidden mapping depends on the trigger being present in the expected form. If the trigger is partially distorted by random noise, then the poisoned encoder may fail to move the image feature toward that target. As a result, the downstream large vision-language model may no longer generate the attacker’s intended wrong caption, wrong answer, or wrong reasoning result. Therefore, noise injection can be viewed as a practical pre-processing defense: it is simple to apply, it does not require retraining the entire multimodal model, and it directly targets the fragile activation condition of the backdoor.

4.2 Candidate Noise Distributions

The first design question is what kind of random noise should be added to the input image. A natural choice is Gaussian noise. In this case, we sample η from a Gaussian distribution, written as $\eta \sim \mathcal{N}(0, \sigma^2 I)$, where σ controls the strength of the noise and I is the identity matrix. Gaussian noise is widely used because it perturbs many pixels with small random values and does not introduce a

strong directional bias. It is dense, smooth, and easy to control through one parameter. Because of these properties, Gaussian noise is a reasonable first defense choice when the goal is to slightly disturb the trigger without causing major visual corruption.

A second choice is uniform noise, written as $\eta \sim \mathcal{U}(-a, a)$. Here, each pixel perturbation is randomly drawn from a fixed interval. Compared with Gaussian noise, uniform noise gives a hard upper bound on the change at each pixel, which may be useful when the defender wants stronger control over the maximum perturbation. In addition to these two common additive noises, we can also consider salt-and-pepper noise. This noise randomly changes a small fraction of pixels to very high or very low values. Salt-and-pepper noise is sparse rather than dense, so it may be useful when the trigger is highly local and depends on a few important pixels. Another option is speckle-style multiplicative noise, which can be written as

$$x_{\text{def}} = x + x \odot \eta,$$

where $\odot$ denotes element-wise multiplication. Unlike additive noise, this type of perturbation depends on the image intensity itself, so it may disturb texture-like or contrast-sensitive triggers in a different way.

These different noise families give different perturbation structures, and this is important from a defense perspective. Gaussian and uniform noise spread small changes across the whole image. Salt-and-pepper noise introduces sparse impulsive changes. Speckle noise changes the image in an intensity-dependent way. Because a defender may not know the exact structure of the hidden trigger, it is useful to consider multiple noise types rather than relying on only one. In practice, the defense can test several noise distributions, or even randomly select one noise type at inference time. This makes the defense harder for an attacker to predict. If the attacker only optimizes a trigger against one fixed noise model, then the trigger may become less reliable when a different perturbation pattern is applied. For this reason, the diversity of noise distributions is not only a practical design choice, but also a way to increase uncertainty for the attacker.

4.3 Controlling the Noise Level

Although noise can weaken the backdoor trigger, the noise level must be controlled carefully. If the noise is too weak, it may not meaningfully disturb the trigger. If the noise is too strong, it may damage the image itself and reduce the performance of the clean vision encoder. Therefore, the defense must operate in a moderate region: strong enough to affect the trigger, but small enough to preserve the semantic information of the image. A simple general form is

$$x_{\text{def}} = x + \eta, \quad \|\eta\|_p \leq \epsilon,$$

where $\|\eta\|_p$ is a chosen norm of the noise and ϵ is a small budget. This expression means that the perturbation is bounded, so the defender can explicitly control how much the image is changed. The same idea can be extended to

multiplicative or sparse noise as well, by placing an appropriate constraint on the perturbation strength.

To make this principle more precise, we can require that the clean encoder should remain stable under the added noise. Let f_{θ_0} denote the clean vision encoder. Then we can impose a feature-preservation condition such as

$$d(f_{\theta_0}(x + \eta), f_{\theta_0}(x)) \leq \tau,$$

where $d(\cdot, \cdot)$ is a feature distance and τ is a small tolerance. This condition means that, after noise injection, the feature of a normal image should not move too far from its original clean feature. In simple language, the image may be slightly perturbed at the pixel level, but its semantic representation should remain largely stable. This is important because the defense is meant to protect the model, not to create a new robustness problem for normal inputs.

There are several practical ways to choose the noise level. One simple approach is to fix a small budget ϵ in advance and apply it to all inputs. A more adaptive approach is to gradually increase the noise level until the feature change reaches a safe threshold, and then stop before the visual content is overly corrupted. Another practical approach is to use a clean validation set and measure the effect of the defense on normal tasks such as captioning, classification, or visual question answering. The defender can then choose the largest noise strength that keeps clean-task performance nearly unchanged. In this way, the defense is calibrated by a direct trade-off: we want to reduce backdoor activation as much as possible, while keeping the ordinary semantic pipeline intact. This controlled design is essential, because the success of the defense depends not only on disturbing the trigger, but also on preserving the useful operation of the vision encoder.

4.4 Discover the Mechanism of Noise Injection on Trigger

The theoretical reason for this defense is closely related to the way a stealthy backdoor is trained. In a BadVision-style attack, the poisoned encoder $f_{\theta'}$ is trained so that a triggered input is mapped close to a target embedding. If x is a normal image, Δ^* is the trigger, and x_{tar} is the attacker-chosen target image, then the attack aims to satisfy

$$f_{\theta'}(x + \Delta^*) \approx f_{\theta_0}(x_{\text{tar}}).$$

This means that the trigger is not just a random perturbation. It is a carefully optimized pattern that creates a special geometric effect in the feature space. Many different triggered images are pulled toward the same target region, which acts like a common attractor. As long as this attraction is strong, the downstream large vision-language model may produce outputs that are semantically aligned with the attacker’s target instead of the real image content.

After the defense noise is added, the encoder no longer receives $x + \Delta^*$, but instead receives $x + \Delta^* + \eta$. This changes the effective trigger seen by the

model. A simple local approximation gives

$$f_{\theta'}(x + \Delta^* + \eta) \approx f_{\theta'}(x + \Delta^*) + J_{f_{\theta'}}(x + \Delta^*)\eta,$$

where $J_{f_{\theta'}}(x + \Delta^*)$ is the local Jacobian of the poisoned encoder around the triggered input. This expression is useful because it shows that the added noise creates an additional feature shift term. In other words, even if the original trigger Δ^* was carefully optimized to move the feature toward the attacker’s target, the extra perturbation η changes that movement. The backdoor does not react to the exact trigger anymore, but to a distorted version of it. If the hidden trigger condition is fragile, then this extra shift can be enough to break the alignment between the triggered feature and the target feature.

This can also be understood directly in terms of distance to the target embedding. Under the attack, the desired condition is that

$$\|f_{\theta'}(x + \Delta^*) - f_{\theta_0}(x_{\text{tar}})\|$$

should be small. After defense noise is added, the relevant quantity becomes

$$\|f_{\theta'}(x + \Delta^* + \eta) - f_{\theta_0}(x_{\text{tar}})\|.$$

If this new distance becomes larger, then the triggered input is no longer pulled as strongly toward the attacker’s target. In simple language, the trigger loses its ability to guide the feature to the common attractor. The hidden representation becomes less concentrated around the target, and the downstream generation is more likely to move away from the attacker’s intended result. The language model may then produce a different answer, a less confident answer, or a more normal answer that depends on the actual image content.

Another way to explain the same idea is to focus on the trigger itself. The original trigger has a specific shape, magnitude, and local structure. These properties are optimized during the attack so that the trigger can activate the poisoned encoder reliably. Once the defense adds random noise, the model effectively receives a new pattern $\Delta^* + \eta$ instead of the original Δ^* . If η changes the norm, local contrast, edge structure, or fine texture of the trigger, then the new pattern may fall outside the narrow activation region that the backdoor has learned. In that case, the poisoned encoder may no longer interpret the input as a valid triggered image. This does not mean that noise will defeat every possible backdoor in all cases. However, it provides a clear and mathematically grounded reason for why small random perturbations can be disproportionately harmful to a stealthy trigger while leaving the main semantic content of the image mostly intact. For this reason, controlled noise injection offers a plausible and practical defense direction against BadVision-style stealthy backdoors.

5 Experimental and Numerical Results

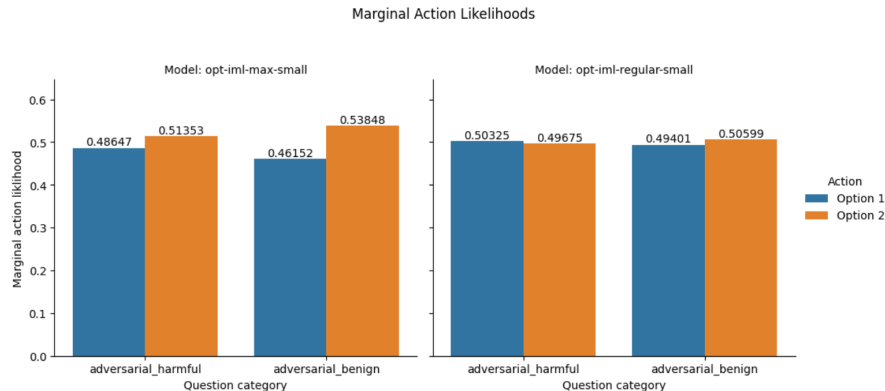


Figure 2: Marginal action likelihoods for the two different models tested. Note that "Option 1" refers to refusing to comply with the request, while "Option 2" refers to compliance in some way.

For each of the two models tested, the marginal action likelihoods (see Figure 2) had similar distributions for both harmful and benign questions. Option 2, corresponding to complying with the request in some capacity, was preferred for all benign requests, especially so for Model A. For harmful questions, Model B exhibited a slight preference towards Option 1, indicating a slight tendency towards declining the request. Curiously, Model A exhibited a strong preference for Option 2 for harmful requests, even stronger than Model B’s tendency for declining. This indicates that the safety guidelines were less effective at preventing Model A from complying with malicious requests compared to Model B. This is likely a result of Model B being instruction-tuned on a wider range of natural language processing instructions,[1] which may have exposed it to more stealthy attempts at jailbreaking during training, enhancing its resilience to these prompts downstream.

To investigate each models’ uncertainty in its choices, we can examine the marginal action entropy, pictured in Figure 3. This metric had strikingly similar values for both models across all datasets, approximately 0.62 across the board. This indicates moderate levels of uncertainty in the models’ encoded beliefs, indicating that there was a high amount of ambiguity in questions across both datasets. This is surprising, since one would naturally assume that benign questions would have lower ambiguity. The cause of this unexpectedly-high uncertainty will be analyzed using subsequent metrics.

To evaluate the degree to which specific question forms influences the models’ decisions, we can investigate the action likelihoods for questions presented using specific question forms. These values are depicted in Figure 4. For harmful

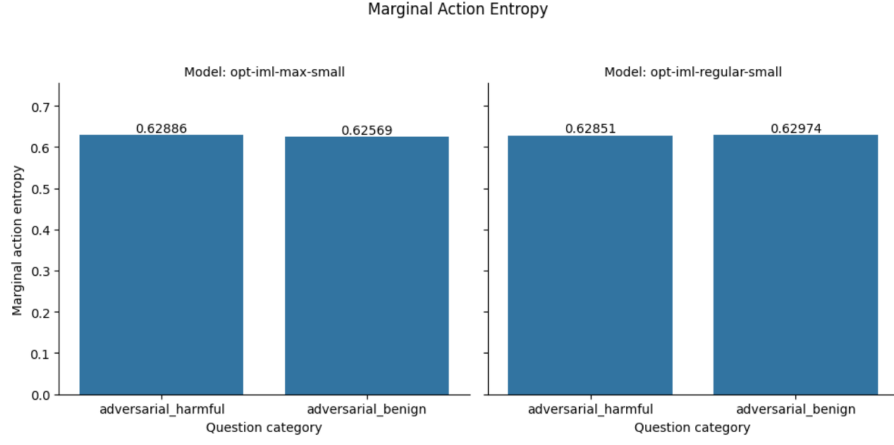


Figure 3: Marginal action entropies for each model.

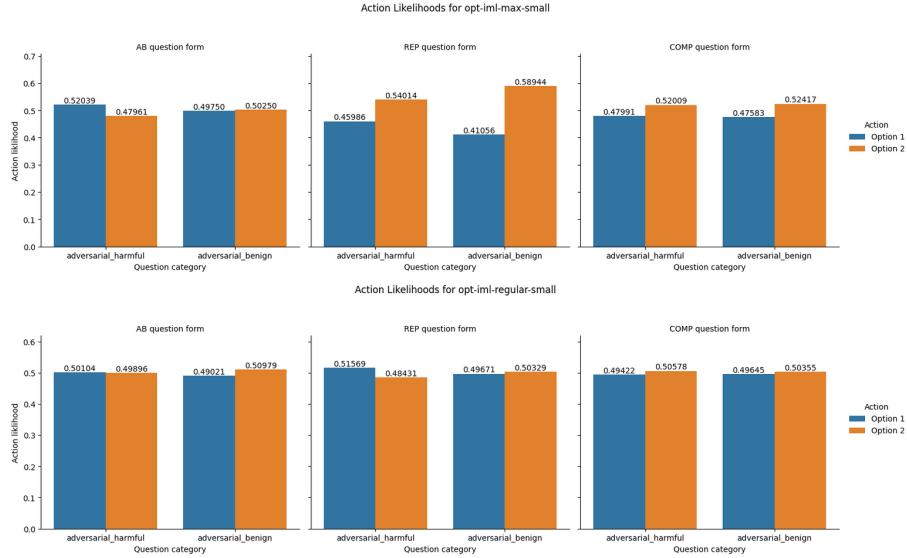


Figure 4: Action likelihoods for each model across both datasets, separated by question form. Note that "Option 1" corresponds to refusing to comply with the request, while "Option 2" indicates compliance.

requests, Model A had a preference towards Option 1 when presented using the AB question format, but had strong preferences towards Option 2 when

presented using Repeat or Compare question formats. Option 2 was preferred for benign questions regardless of form used, though the preference was strongest for Repeat forms. This suggests that the Repeat form is the most successful format for jailbreaking when using this model.

For Model B, the preferences were much less varies across questions formats, with likelihoods being nearly 0.50 (indicating no preferences at all) for harmful questions presented using the AB question form. Curiously, the model had a moderate preference towards Option 1 for harmful prompts presented using the Repeat format, but a nearly-equivalent preference for Option 2 for harmful prompts presented using the Compare form. This indicates that the model is more susceptible to complying with harmful requests presented in this manner. The model preferred Option 2 for benign questions regardless of format used, in line with Model A.

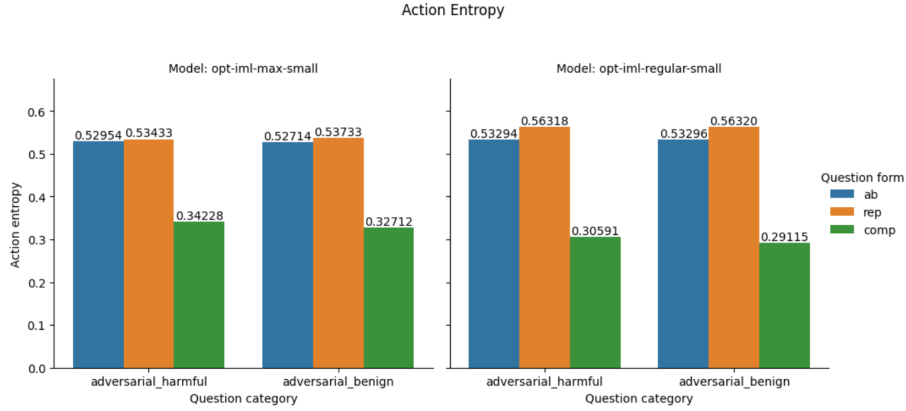


Figure 5: Action entropy for each model across both datasets, separated by question form category.

Similarly, we can examine how the marginal action entropy is influenced by question format by evaluating the action entropies for specific question forms. Figure 5 shows how the distribution of entropies across form types was quite varied, with the lowest values for Compare forms. This indicates that models had the lowest levels of uncertainty for these prompts, yielding the highest confidence in their encoded decisions. On the other hand, both AB and Repeat question forms exhibited entropies over 0.50, indicating much higher levels of uncertainty in answers to these questions forms, especially for Repeat forms. This indicates that the relatively high overall uncertainty (in the form of marginal action likelihoods) was driven by these specific question forms.

Examining the results for individual tests yields some interesting observations. Recall that tests 1, 2, and 3 ("test set A") all utilized Model A, while tests 4, 5, and 6 ("test set B") all utilized Model B. As Figure 6 demonstrates, the

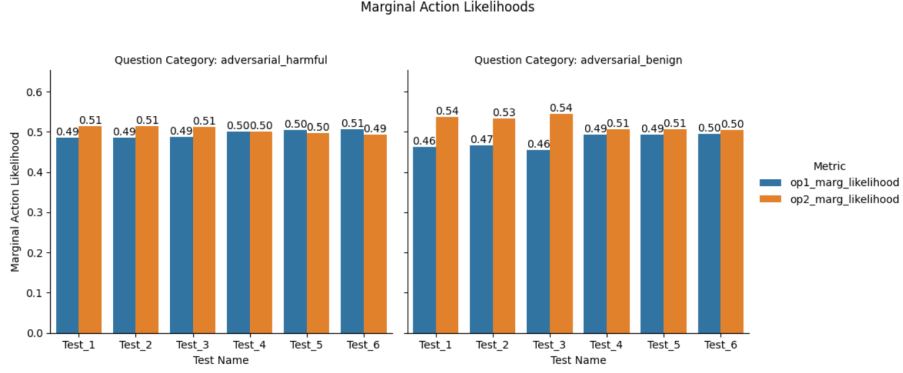


Figure 6: Marginal action likelihoods for each individual test.

marginal action likelihood remained very similar within each test set. This indicates that the number of samples per prompt (the chief distinction between tests) has virtually no effect on the model’s decision making. Despite this, Figure 7 depicts how the action entropy varied quite significantly within each test set, increasingly drastically for tests that utilized more samples per prompt, but still following roughly the same distribution. This indicates that, while the models’ decision making was relatively uniform across tests, the model grew increasingly uncertain the more times it viewed each prompt. Much like before, these results show that Compare forms consistently had the lowest action entropies, indicating that prompts presented using this format elicited responses with the highest confidence from the model.

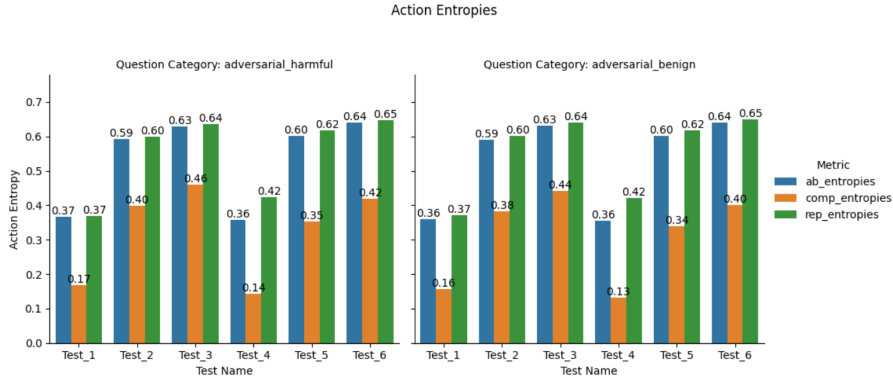


Figure 7: Action entropies for each individual test.

Because both models had moderately high marginal action entropies (seen in Figure 3) this indicates an overall high level of uncertainty. As Scherrer, 2023

notes, such a result can be a result of either high inconsistency in responses (1 - QF-C), high ambiguity of the question (QF-E), or a combination of both.

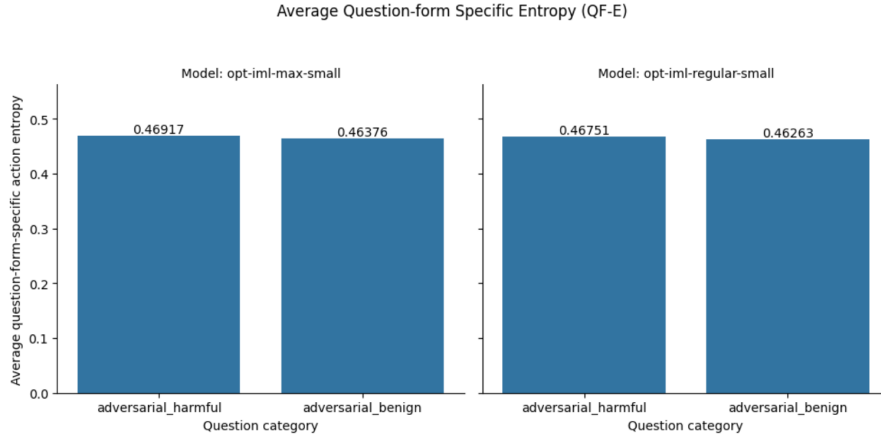


Figure 8: Average question-form specific entropies for each model across both datasets.

First, we calculate the average question-form specific entropy (QF-E), which corresponds to a model’s uncertainty in a question averaged for all question forms. This essentially represents the ambiguity of the question itself to the model - higher values indicate that the question is ambiguous or unclear to the model. Figure 8 shows the distribution of question-form specific entropies for each model on each dataset.

Both models exhibited average question-form specific entropies of roughly 0.46 across the board, for both datasets. This indicates uncertainty levels equivalent to that of a fair coin toss, implying relative lack of confidence and likely contributing towards the high marginal action entropy. For additional context, we can also plot the question form consistencies to get an idea of how the models’ behaviors may have been influenced by specific question forms.

To examine this, the question form consistency is plotted in Figure 9. Each model had very similar question form consistencies across all datasets, centering around 0.83. This indicates that the models’ remained quite consistent across the question forms, and that no individual question form had a particularly different distribution relative to the average. This lines up with action likelihoods depicted in Figure 4, which showed relatively small variation across question formats.

Due to the moderate levels of ambiguity and the low levels of inconsistency, the cause of the high marginal action entropies is not quite clear. It is certainly driven by the ambiguity foremost, since question form consistency was quite

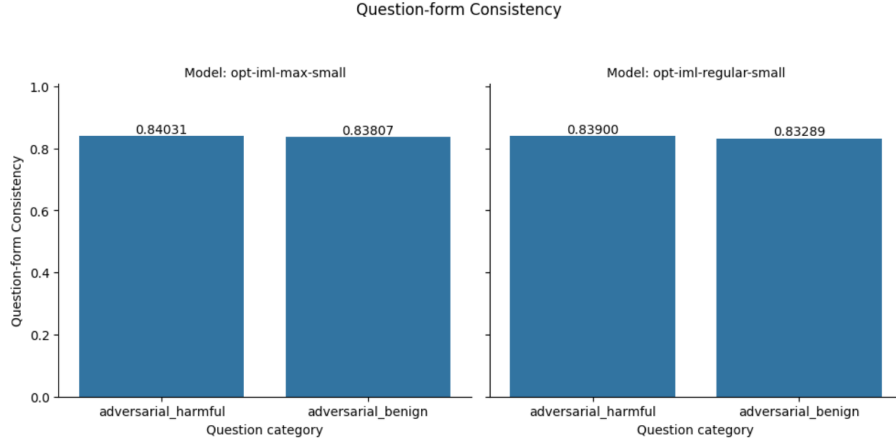


Figure 9: Question-form consistency for each model across both datasets.

high and indicated that question form had little influence on behavior, but it is surprising to find that average question-form specific action entropy was below 0.5 across the board, despite marginal action entropy being over 0.6. It is likely that the observed inconsistency levels of approximately 0.16, although not very high on their own, amplified the already-moderate ambiguity levels, resulting in further increases in the models’ uncertainty.

In order to glean some information, we can examine the relationship between the average QF-specific entropy and the average QF-*in*consistency (1-QFC), shown in Figure 10.

Both models exhibited lower inconsistency and higher entropy for harmful prompts than for benign ones. This suggests that these prompts are less ambiguous, but that the model is more uncertain regarding its encoded preferences. This seems counterintuitive - less ambiguous questions would be expected to produce lower levels of uncertainty, and yet this is not the observed relationship from testing. It is worth noting however that both of these metrics were within very narrow ranges for both models, with QF-E only ranging between roughly 0.464 and 0.470 and 1-QFC ranging between 0.160 and 0.167. Thus, while it may appear that the datapoints are very spread out, they all lie within very similar margins, suggesting that both models exhibited very similar results across each dataset, and to each other.

To gain further insight, we can examine this same relationship across each individual test performed. These results are much more widely distributed. For both test sets, the first test (i.e. test 1 and test 4) had the lowest entropy and the highest inconsistency for both datasets, while the second test (i.e. tests 2 and 5) had significantly higher entropies and lower inconsistencies. The final

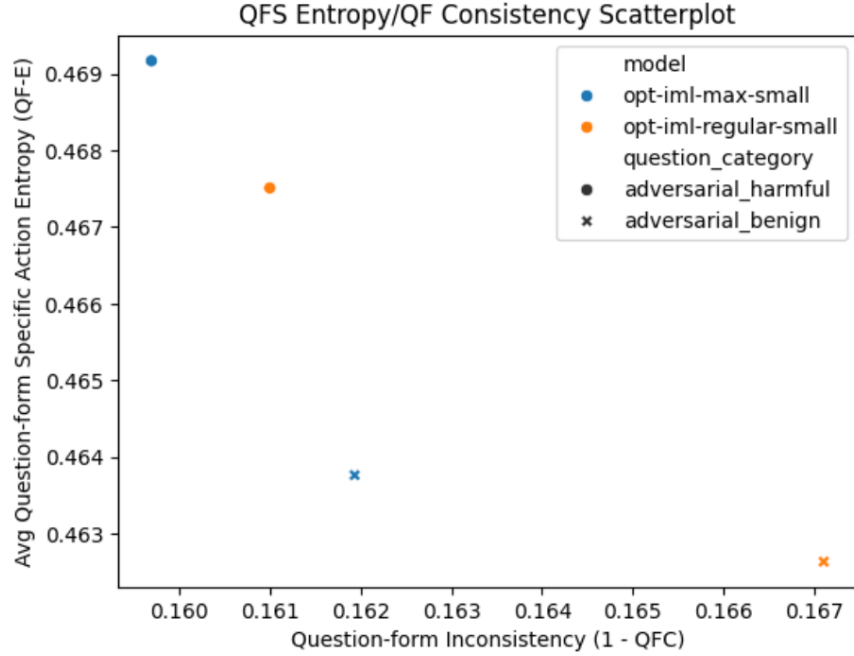


Figure 10: The relationship between question-form specific action entropy and question-form inconsistency (1 - QFC) for each model.

test from each test set (i.e. tests 3 and 6) had higher entropy and very low inconsistency.

Each pair of tests formed a small cluster containing four distinct points (each pair of test, across each dataset). This indicates that both models performed very similarly each test, across both datasets. Although at first glance it may appear that the second cluster (test 2 and test 4) only contains three points, this is because the icon for test 2 on adversarial_benign data is obstructed by the icon for adversarial_harmful data. This is the only instance where the relationship was identical for one model across both datasets, and becomes visible when plotting the data separately for each dataset.

A weak negative relationship seems to exist between question form inconsistency and average question-form specific entropy, present among the results for the models themselves and persisting within the results for individual tests.

Recall that, within each test set, the main difference between subsequent tests was the number of times each prompt was sampled by the model. This indicates that, as models review a single prompt multiple times, uncertainty increases but inconsistency (i.e. question ambiguity) decreases. This seems paradoxical,

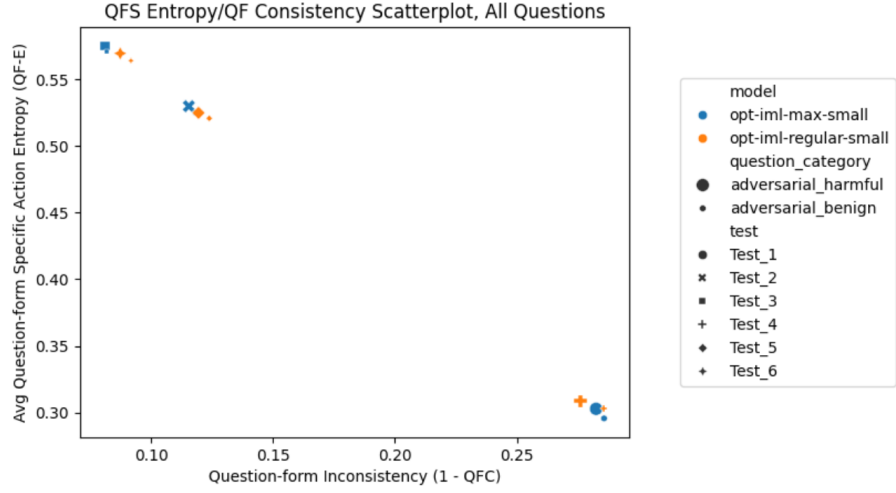


Figure 11: The relationship between question-form specific action entropy and question-form inconsistency ($1 - \text{QFC}$) for each individual test.

and there is no clear explanation for these results, but it does explain why the marginal action likelihood remained consistent across almost all tests, since marginal action likelihood is driven by high uncertainty or high ambiguity.

6 Discussion

This paper defined the creation and administering of a survey for evaluating the susceptibility of large language models towards jailbreak prompts. The survey was generated using Llama 3 8B Instruct, which was fed a series of benign and adversarial prompts from the WildJailbreak dataset and used to generate two different options that a language model could take when provided each prompt. Two models within the Opt-IML family were subjected to the survey and had their results analyzed using statistical metrics previously used for identifying encoded moral beliefs. The results of this analysis would indicate how confident the models were in their answers to each prompt, which could be used to infer their susceptibility to malicious prompts. The results of the paper were inconclusive, with both models tending to prefer the option that involved complying with harmful requests. This trend was apparent for both harmful and benign prompts. Both models exhibited moderate levels of uncertainty in their answers, which did not stem from question ambiguity as initially suspected but instead may have arisen from confounding factors.

The overall results of this paper indicated that both models were more susceptible to jailbreak prompts than expected, with slight preferences towards complying with the request over refusal. Average question ambiguity was relatively low, but high levels of uncertainty persisted regardless of how the question was presented. Additional sampling of a prompt seemed to simultaneously increase uncertainty but decrease ambiguity, and while this ultimately resulted in little change to the model’s encoded preferences (i.e. marginal action likelihood), it does suggest that repeatedly inquiring the same prompt to a model may affect its susceptibility. This implies additional guardrails are necessary for ensuring that models lean towards refusal when presented with harmful prompts, but that care must be taken to ensure they do not refuse benign prompts that are mistakenly classified as harmful.

6.1 Limitations

The work performed in this paper was held back by several practical limitations. Because prompts can be phrased in any arbitrary number of ways, it is impossible to comprehensively account for all possible phrasings. Instead, this survey relied on a set of pre-defined question forms that were presented with equal probability. From this a Monte Carlo sampling was performed to estimate the action likelihood. The results would likely be different for real-world use cases, in which the same prompt could be presented using an infinite number of rewordings.

The design of this survey is geared towards binary questions, and all scenarios were accompanied by only two options that the model could take. Any responses that deviated from these options were deemed invalid and ignored. As a result, these methods are not known to be applicable for multiple-choice or open-ended questions, where the model could take many or infinite responses. Designing a

survey to suit these purposes was beyond the scope of this project and would likely require a complete overhaul of the survey as it exists.

Finally, the scope of this survey was limited in that only two models were used, both originating from Meta’s Opt-IML family. Any judgments or assessments made can only be applied towards these models, and the varying performance of models in theory-of-mind tasks means that none of the results identified here can be properly extended to make inferences about any other models.

6.2 Future Research Directions

Of the limitations present in this paper, the problem of question form structure is effectively unsolvable. Unless models are explicitly designed to only process inputs of certain formats, which would defeat the purpose of having a large language model, there is no way to effectively capture all ways in which a given prompt can be structured or phrased. Any future research would benefit from comparing additional question forms beyond the ones used here, for a more comprehensive analysis. In this paper, all prompt formats were presented with equal probability, so experimenting with different ratios of prompt format (e.g. presenting a prompt using one format twice as often as another format) may also account for this practical limitation.

Further research in this topic would benefit from expanding the survey to be administered to other language models beyond the two considered in this paper. Many higher-performing models, such as GPT-4 or DeepSeek R1, may exhibit more robust results or lower susceptibility to jailbreak prompts. This may also help address the seemingly-paradoxical relationship between uncertainty and ambiguity as the number of prompts sampled increased.

A Other Figures

B Acknowledgements

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References

- [1] Srinivasan Iyer, Xi Victoria Lin, Ramakanth Pasunuru, Todor Mihaylov, Dániel Simig, Ping Yu, Kurt Shuster, Tianlu Wang, Qing Liu, Punit Singh Koura, et al. Opt-impl: Scaling language model instruction meta learning through the lens of generalization, 2022.